How Foveated Vision Shapes Cognitive Maps in Navigation Agents

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Abstract

What shapes the content of a recurrent navigation agent's learned internal state — the task, the architecture, or the structure of the visual input? Prior work has shown that blind PointNav agents spontaneously develop cognitive-map-like representations in their recurrent state under task pressure alone [Wijmans et al., 2023]. We show that the blind result is a special case of a broader principle: *the visual encoder's information capacity determines whether the LSTM has to carry world-frame spatial structure*. Holding task, optimiser, and LSTM backbone identical and varying only the visual pathway (no vision; a resolution-collapsed encoder; uniform vision; foveated vision at a fixed gaze; foveated vision with a learned gaze), we probe the recurrent state for world-frame position and heading. All five conditions encode top-layer GPS in their LSTM in the typical-episode window. What separates conditions is what happens *later*: when the encoder hands the LSTM minimal spatial-feature variety per step (blind: no encoder; matched-compute: $1{\times}1$ feature map), the LSTM *maintains* a persistent, integration-capable code across episode duration; when the encoder hands it richer spatial features (full-resolution encoders, $8{\times}8$ feature map), the LSTM *overwrites* the GPS code as those visual features take over. We call this the *encoder–memory race*; it recasts the "cognitive maps emerge without vision" story as "cognitive maps emerge — and persist — when the visual pathway cannot resolve the map on its own." A second finding cuts across conditions: five agents that solve the task equally well develop recurrent representations in essentially orthogonal linear subspaces, and cross-condition memory transplants incur task cost well above transplant mechanics — sensor structure shapes representational *format*, not only strength. A third question — whether static gaze location within a rich-encoder regime acts as a second content axis — is tested by a foveated-shifted causal control whose training is in progress. The H1 ordering reproduces on a held-out MP3D evaluation, consistent with an agent-internal rather than Gibson-specific mechanism. Three implications, each scoped to the recurrent PointNav setting we study. **(i)** Encoder capacity correlates inversely with linear, top-layer LSTM spatial encoding; *if* the correlation reflects causation (which the encoder-resolution scaling sweep, in progress, tests directly), it would also be a practical training-time lever for inducing cognitive-map-like internal representations. **(ii)** "More pixels, richer memory" is the wrong intuition within this setup; the relationship across our five conditions is inverse. **(iii)** The pattern parallels independent animal-navigation findings — enhanced hippocampal coupling in congenitally blind humans, the disruption of rodent grid-cell firing in the dark (with navigation nonetheless preserved), cross-modality hippocampal remapping in bats — without depending on their biophysics, suggestive of a sensor–memory coupling principle whose generality across architectures (transformers, non-RL learners, non-navigation tasks) we do not test here.

Preprint.

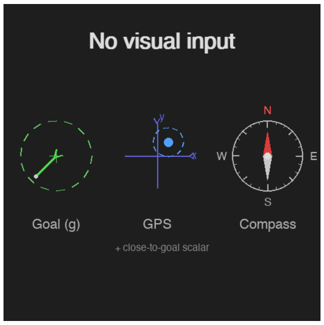
1 Introduction

Recurrent navigation agents trained with deep RL build internal states that support memory, planning, and generalisation, and the content of those internal states is at least as interesting as the task metrics they drive. A natural question is what *determines* that content: architecture, objective, training budget, or visual input? Wijmans et al. [Wijmans et al., 2023] showed that map-like spatial representations emerge from task pressure alone, even in blind agents that receive only goal displacement and proprioception; Dwivedi et al. [Dwivedi et al., 2022] probed sighted agents for related concepts; Banino et al. [Banino et al., 2018] and Cueva & Wei [Cueva and Wei, 2018] demonstrated grid-cell-like structure in path-integration networks. Together these results establish that *some* spatial structure is learned without being designed in. They leave open the complementary question that we take up here: within agents that all achieve high task success, how much of *what* recurrent memory encodes is determined by the structure of the visual input rather than by the task? In biology the answer is clear — primate hippocampal cells code joint place–view–action state because primates sample the scene with a foveated sensor [Wirth et al., 2017, Rolls, 2024], and the same bat produces non-overlapping hippocampal populations in vision versus echolocation [Geva-Sagiv et al., 2016]. We investigate the analogous question in a controlled agent setting where the only free variables are the visual input structure and the gaze that directs it.

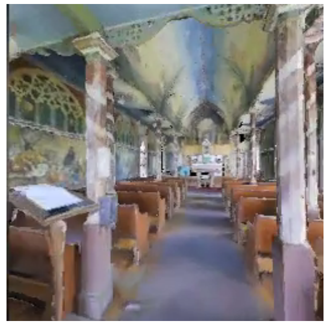
Our five-condition design isolates three degrees of freedom: sensor type (blind / uniform / foveated), encoder capacity (uniform's full ResNet-18 vs. matched-compute's 1×1-collapsed feature map), and gaze location (fixed vs. learned). Every condition uses the same DD-PPO learner, the same Gibson-0+MP3D training pool, the same LSTM backbone, and the same non-visual sensor stack (Figure 1). We adopt the LSTM architecture from Wijmans et al. [Wijmans et al., 2023] deliberately: the emergent-map literature we build on is recurrent, and matching that architecture lets our findings be interpretable as statements about *the structure of the task-shaped memory* rather than about the recurrent-versus-transformer design choice. We discuss in §5 whether the logic plausibly extends to transformer-backed or non-navigation systems — an empirical question we do not settle here.

We test three hypotheses. **H1 (encoder–memory race).** Agents whose visual encoder cannot resolve world-frame position or heading should develop these quantities in their recurrent state; agents with a rich visual encoder should not. Signature: current-state GPS / compass probe R^2 tracks visual-encoder information content monotonically across the five conditions, with persistent decoding at lags of ≥ 10 time-steps. **H2 (sensor structure $\rightarrow$ format divergence).** Different sensor conditions should produce internal representations in distinct linear subspaces, not scaled versions of a common code. Signature: pairwise CKA near zero, probe weights fail to transfer across conditions while within-condition probes succeed, and cross-condition memory transplants degrade task performance beyond same-condition self-transplants. **H3 (gaze location $\rightarrow$ orthogonal content lever).** Where a foveated sensor is centred should act as a representational-content axis independent of the sensor transform. We test this with a foveated-shifted control in which gaze is hardcoded to a different image location from the default central gaze; same foveation transform, same training budget, different gaze target. Our minimal learned-gaze module collapsed to a static output (Appendix B), so we use that collapsed location for the shifted-gaze control rather than as evidence about active gaze; the foveated-shifted training is in progress (§4.5). Each hypothesis has a direct analog in the animal-navigation literature [Kupers and Ptito, 2014, Chen et al., 2016, Geva-Sagiv et al., 2016, Jutras et al., 2013, Najemnik and Geisler, 2005], which we return to in Discussion as evidence that the effects reflect general sensor–memory coupling principles rather than architectural artefacts.

Findings H1 is supported by a clean monotonic effect at the level of *temporal stability* of the top-layer GPS code. All five conditions encode GPS in the LSTM in the typical-episode window (steps 50–200, $R^2 \in [0.6, 0.95]$); the bottleneck conditions then *maintain* that code across episode duration (blind $R^2 = 0.92$ even at step 800+; matched $R^2 = 0.68$), while rich-encoder conditions *overwrite* it as visual-encoder features take over (uniform / foveated / foveated-learned all collapse to chance or below by step 400–800). Aggregated across full-length episodes (Table 1, no step cap), this presents as $R^2 = 0.95/0.78$ for blind / matched-compute and $R^2 \in [-2.4, +0.06]$ for the rich-encoder conditions. Bottleneck conditions' code is persistent at lags ≥ 20 time-steps. The bottleneck-vs-pass-through partition is preserved on a held-out MP3D evaluation, consistent with an agent-internal rather than Gibson-specific mechanism. H2 is supported by four independent measures of cross-condition divergence: pairwise CKA at the noise floor ($< 10^{-4}$), catastrophic probe-transfer failure ($R^2 \ll -800$ in every off-diagonal cell), perfect 1-NN cluster purity (1.000 across 7500

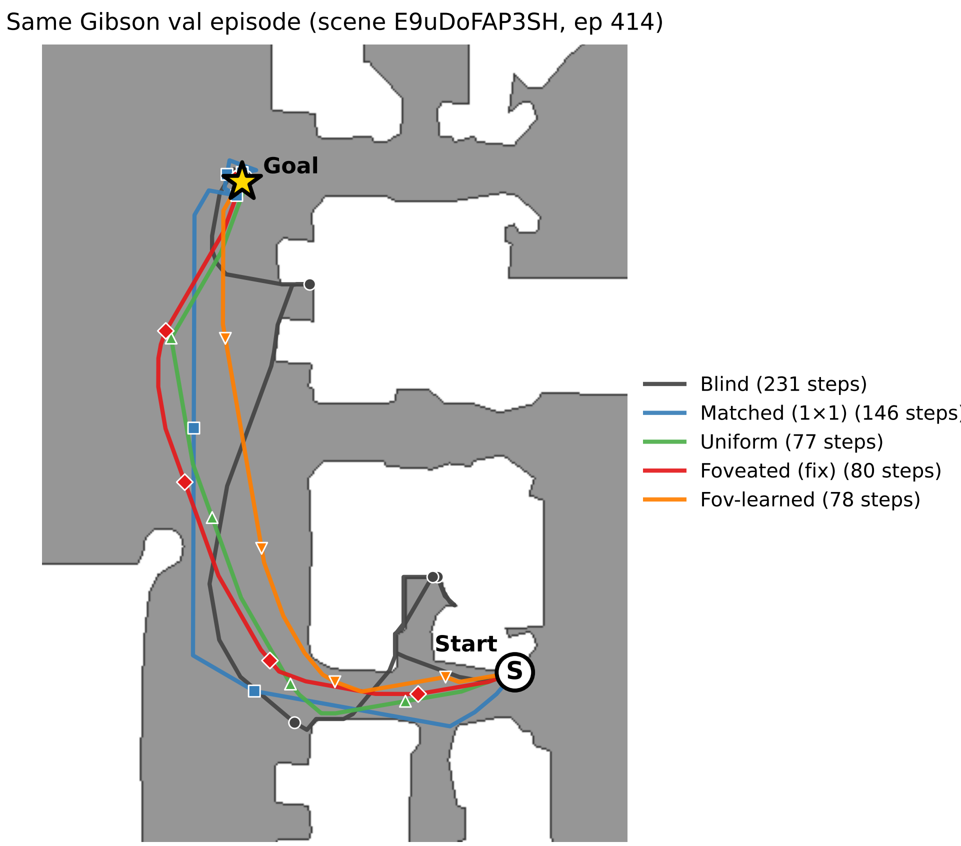


(a) Blind



(b) Uniform

(c) Foveated (fixed gaze)



(d) The 5 conditions on the same MP3D-train episode (scene E9uDoFAP3SH, ep 40422 in the env iterator; same paired index 414 across our probing data), trajectories overlaid on the Habitat-rendered top-down occupancy map. Each condition's SPL on this episode is within ± 0.10 of its per-condition median, so all five are typical successful runs. Bottleneck conditions (blind 231 steps, matched 146) take longer paths through the corridor; rich-encoder conditions (uniform / foveated / fov-learned, 77–80 steps) take near-direct paths along the same corridor.

Figure 1: The five visual conditions differ only in what reaches the agent's encoder, but produce visibly different navigation behaviours. *Inputs:* (a) Blind: no visual input; only the non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar) — the Wijmans [Wijmans et al., 2023] configuration. (b) Uniform: 256×256 full-resolution RGB. (c) Foveated (fixed gaze): eccentricity-dependent Gaussian blur centred on the image middle ($\sigma_{\max} = 8$, quadratic falloff). Two additional conditions not pictured: *matched-compute* (a down-sampled uniform view at 48×48, whose total pixel budget matches foveated and whose ResNet-18 encoder produces a 1×1 feature map) and *foveated (learned gaze)*, which adds a learned (x, y) gaze offset to (c). *Behaviour:* (d) all five conditions solve the same PointGoal navigation task; the trajectory shape and length differ systematically, previewing the encoder–memory race finding (§4.2).

pooled hidden states, vs. chance 0.20), and behaviourally costly cross-condition memory transplants (SPL drops 0.19–0.21 between rich-encoder pairs, well above the self-transplant noise floor). H3 asks whether gaze location is a second content axis *within* the rich-encoder regime; the clean causal test (foveated-shifted control) is in training at submission and will populate §4.5. A non-linear (MLP) probe recovers position from the rich-encoder *top-layer* hidden states ($R^2 \in [0.51, 0.73]$), sharpening rather than weakening H1. We interpret this together with the per-layer probes (Figure 2c): every condition has linearly-decodable GPS at LSTM Layer 0 (where the GPS sensor embedding enters the input vector — so this is at least partly a sensor-pass-through effect, not a network "computation"); the divergence is what happens through Layers 1–2. Bottleneck conditions retain linearly-accessible GPS to the top layer used by DD-PPO's linear action head; rich-encoder conditions do not. The MLP recovery at the top layer is consistent with position information persisting in a non-linear subspace, though we cannot from this alone rule out MLP exploitation of position-correlated features (wall distance, scene identifiers) rather than position itself — a question the encoder-resolution scaling sweep (Appendix E, in progress) will help resolve. H1 is most precisely stated as a claim about *linear, top-layer, policy-accessible* GPS encoding.

Contributions (i) A controlled five-condition ablation that decouples "no vision" from "visual-encoder bottleneck" via the matched-compute condition, making the encoder–memory race claim falsifiable. (ii) Convergent evidence (linear probes, MLP probes, CKA, probe transfer, 1-NN purity, memory transplant, shortcut discovery, MP3D generalisation, per-layer probes) for two of the three hypotheses, with a clean methodology disclosure including the deterministic-rollout protocol that fixes a probe-collection confound documented in §3.3. (iii) A mechanistic re-reading of Wijmans et al.'s blind-agent finding: cognitive maps emerge not because vision is absent but because the visual encoder cannot resolve the map, predicting an encoder-capacity scaling curve we begin to test in Appendix E.

2 Related Work

Cognitive maps in RL navigation agents Wijmans et al. [Wijmans et al., 2023] demonstrated that map-like representations emerge in the recurrent memories of DD-PPO agents even without visual input; Dwivedi et al. [Dwivedi et al., 2022] probed sighted agents for related spatial concepts; Banino et al. [Banino et al., 2018] and Cueva & Wei [Cueva and Wei, 2018] found grid-cell-like units in recurrent networks trained on path integration; Gornet & Thomson [Gornet and Thomson, 2024] extended this via visual predictive coding. Hennig et al. [Hennig et al., 2023] showed that RL under partial observability can induce belief-state-like representations without explicit probabilistic objectives. Ramakrishnan et al. [Ramakrishnan et al., 2025] examined spatial cognition in frontier models. All these studies use either absent or spatially uniform perception; none vary input quality spatially, and none test whether different sensors produce different representational formats.

Foveated vision and gaze policies in deep learning Foveated rendering [Strasburger et al., 2011, Rosenholtz, 2016] exploits eccentricity-dependent acuity falloff for efficiency. Learned glimpse policies appear in recurrent attention models [Mnih et al., 2014]; Atanov et al. [Atanov et al., 2024] showed that sensor layout changes downstream task performance. Pourrahimi & Bashivan [Pourrahimi and Bashivan, 2025] probed a foveated visual-search model for neuroscience-predicted features; Shang & Ryoo [Shang and Ryoo, 2023] jointly learned gaze and motor policies following the Sprague & Ballard [Sprague and Ballard, 2003] framework. These address visual search or task performance; none asks how sensor structure reshapes the learned spatial memory of a downstream navigation policy.

Probing neural representations Belinkov [Belinkov, 2022] surveys probing classifiers; Hewitt & Liang [Hewitt and Liang, 2019] introduced control tasks to distinguish genuine structure from probe memorisation; Kornblith et al. [Kornblith et al., 2019] introduced linear CKA for cross-model similarity. We adopt linear probing, the Hewitt–Liang selectivity metric, and unaligned linear CKA, and complement them with two behavioural interventions directly on the recurrent state (memory transplant and shortcut discovery) to validate the probe-based findings outside the linear-probe framework.

Biological precedent Our hypotheses and framing draw on three threads of animal-navigation research. (i) Primate hippocampal cells code joint place–view–action state [Wirth et al., 2017, Rolls, 2023]; Rolls [Rolls, 2024] argues that primate *spatial-view cells* and rodent *place cells* reflect the different sampling statistics of foveated versus near-panoramic vision. (ii) The same bat in the same environment produces non-overlapping hippocampal populations when it switches between vision and echolocation [Geva-Sagiv et al., 2016]—format-level divergence across sensor modalities within a single animal. (iii) In primates, each saccade resets hippocampal 3–8 Hz rhythms and the reliability of the reset predicts subsequent memory [Jutras et al., 2013]; only overt (not covert) attention automatically writes to visual working memory [Tas et al., 2016]. Gaze targets themselves are selected in an information-gain-maximising fashion consistent with ideal-observer predictions [Najemnik and Geisler, 2005, Gottlieb et al., 2013, Hayhoe and Matthis, 2018]. We use these results only to frame our ML predictions and interpret our findings; our contribution is the controlled ablation in a learned agent that cross-species and cross-modality biological comparisons cannot cleanly supply.

What is new here Three things distinguish this work from the studies above. (i) A *controlled* five-condition ablation with everything but the visual pathway held identical, including the matched-compute condition that decouples "no vision" from "visual encoder bottleneck" — the lever that makes the encoder–memory race claim falsifiable. (ii) A protocol-corrected probing methodology: deterministic action selection at probe time (the policy used by every other component of our eval pipeline) rather than stochastic sampling, which we found inflates probe R^2 for high-entropy policies via a 4-step quasi-static-trajectory artefact (§3.3, Sampling protocol). (iii) Convergent behavioural validation of the probe-based H1 and H2 findings via memory transplant and shortcut discovery, both of which sit *outside* the linear-probe framework.

3 Methods

3.1 Agent Architecture and Training

All agents are trained on PointGoal navigation in Habitat [Savva et al., 2019] using DD-PPO [Wijmans et al., 2020] on a merged pool of Gibson-0+ [Xia et al., 2018] (411 scenes) and Matterport3D train (61 scenes), totalling 472 training scenes. Evaluation uses 18 held-out MP3D test scenes (1008 episodes).

All agents share an identical 3-layer LSTM (512-d) backbone and the Wijmans non-visual sensor stack: goal-in-start-frame, GPS, compass, and a close-to-goal scalar, each projected to 32-d and concatenated with a 32-d previous-action embedding. All sighted conditions are trained for a target of 250 M environment frames; the blind baseline is trained to 342 M frames to allow slower convergence from proprioception alone.

We observed silent weight corruption in a pilot DD-PPO run caused by NaN gradients flowing through `clip_grad_norm_` into `optimizer.step()`; we mitigate this with a gradient-sanitisation patch to `PPO.before_step`, described in Appendix C.

3.2 Visual Conditions

Five conditions vary only the visual input (Figure 1):

1. *Blind:* no visual input; non-visual sensor stack only, replicating Wijmans et al. [Wijmans et al., 2023].
2. *Uniform:* full-resolution 256×256 RGB encoded by ResNet-18.
3. *Foveated (fixed):* eccentricity-dependent Gaussian blur ($\sigma_{\max} = 8$, quadratic falloff) with gaze locked at image centre; same ResNet-18 encoder.
4. *Matched-compute:* 48×48 uniform RGB. At this resolution, ResNet-18's 32-fold spatial downsampling collapses the feature map to 1×1, removing the spatial dimensions of the encoder's output (channel information is preserved as a flat 512-d vector). **[TODO: Verify whether channel info alone in matched-compute encodes any usable position signal — a small probe on the encoder feature map (not LSTM state) would settle this.]**
5. *Foveated (learned):* same blur model, but gaze centre predicted by a lightweight MLP trained end-to-end with the navigation policy.

The matched-compute condition is central to H1: its 1×1 encoder feature map provides a visual observation while leaving the encoder unable to resolve world-frame position from the current frame, separating the "no vision" account from the "encoder bottleneck" account. We attribute its strong LSTM GPS encoding to the feature-map collapse specifically, but the 48×48 input also has lower spatial resolution than uniform's 256×256, so we cannot from this single condition fully separate "1×1 feature-map collapse" from "low input resolution." The encoder-resolution scaling sweep (Appendix E, in progress) varies input resolution at fixed encoder stack to test this attribution directly.

Implementation note The foveated conditions (fixed, shifted, learned) disable the running-mean-and-variance input normaliser used by uniform and matched-compute, to avoid an in-place autograd conflict along the gaze-decoder gradient path. All reported results were obtained with this setting; we have not directly verified that re-enabling the normaliser would leave the H1 ordering unchanged. **[TODO: Verify normaliser invariance: rerun foveated-fixed with the normaliser re-enabled (the autograd conflict is gaze-decoder-specific, so feasible for fov-fix), and confirm the H1 GPS R^2 does not move.]**

3.3 Probing Methodology

After training, we freeze all weights and collect per-step LSTM hidden states (top-layer $\mathbf{h}$, 512-d) paired with ground-truth spatial labels from $n=500$ Gibson held-out validation episodes (the split not used in training). Rollouts use *deterministic* (argmax) action selection, matching the eval protocol of our shortcut-discovery, memory-transplant, and standard habitat-baselines evaluators. Under this protocol the rollout lengths span roughly $\sim$100–400 mean steps depending on condition (longer for blind, shorter for high-SPL conditions), so we use 5-fold episode-level CV to keep the train/test splits balanced per scene rather than length-matching across conditions. **[TODO: A controlled length-matching ablation (truncate every condition to a common length, re-run probes) is left for follow-up; we expect it to leave the H1 ordering unchanged because the bottleneck conditions decode at high R^2 even at the smallest sub-sample, but have not verified it.]** We train Ridge regression probes ($\alpha = 10$) with episode-level 5-fold cross-validation, reporting mean $\pm\sigma$ across folds, and verify probe specificity with Hewitt–Liang label-permutation selectivity [Hewitt and Liang, 2019].

Probed variables: GPS position (episodic), compass heading (episodic), distance-to-goal, lag-k path history (target value at $t-k$ decoded from $\mathbf{h}_t$, $k \in \{0,\dots,20\}$), visited-region occupancy, full occupancy grid (20×20), per-unit spatial information (place-cell analysis), and the ego-frame goal vector $\mathbf{R}(-\theta)(\mathbf{g}-\mathbf{p})$ decomposed into scalar goal-distance and 2-D direction.

Sampling protocol An earlier draft of this work used stochastic (policy-distribution) action selection for probe rollouts. Under stochastic sampling, conditions with higher action-entropy (specifically the rich-encoder conditions foveated, uniform, matched, blind under their trained stochastic policies) sampled the STOP action early with probability ~0.25 per step, producing mean episode lengths of ~4 steps and target variance two orders of magnitude below the episodic range. The resulting probe R^2 values were trivially high across all conditions (any predictor fits a near-constant target). This confounded cross-condition comparisons, and we switched to the deterministic protocol used here and elsewhere in the probe/eval pipeline. All results in this paper use the deterministic protocol.

For foveated-fixed, all reported analyses (probes, behavioural, eval) use checkpoint ckpt.36 at $\sim$174M frames, the last clean intermediate before the silent-corruption window (Appendix C).

3.4 Cross-Condition Analysis

To test H2, we employ two complementary methods. *Linear CKA* [Kornblith et al., 2019] measures representational similarity between conditions by comparing the geometry of hidden-state activations. Values near 1 indicate identical geometry; near 0 indicates unrelated representations. *Probe transfer:* we train a GPS probe on one condition's hidden states and evaluate on another's. If spatial information is encoded in a shared linear subspace, cross-condition transfer should succeed. Catastrophic failure indicates fundamentally different encoding formats. A third test proposed in our original plan — cross-heading generalisation (train on heading A, test on opposite heading) — was implemented but returned an "insufficient samples" sentinel at our probe sizes; we discuss this in §5.5.

3.5 Behavioural Probes

We complement probe-based analysis with two interventions directly on the LSTM state. *Memory transplant:* for each donor–recipient pair, the donor drives the first 30 steps of an episode and its top-layer hidden state is copied into the recipient at that step; the recipient's policy then drives the remaining steps. We compare *baseline* (recipient alone, full episode), *self-transplant* (recipient as its own donor, controls for transplant mechanics), and *cross-transplant* (different donor). Midpoint 30 was chosen because Gibson PointNav episodes average $\sim$100–120 steps; a later midpoint left too few steps after transplant. We use 150 episodes per pair per condition, sampled uniformly from the held-out Gibson validation split. *Shortcut discovery:* for each condition, we run episodes in pairs of two using the same scene but different goals, sampling 20 scenes with 10 paired episodes per scene (200 paired runs per condition per setting). Per-condition SPL is reported as the mean of the 20 scene-level mean SPLs (prevents high-episode-count scenes from dominating). We compare *reset* (LSTM zeroed between the two episodes) vs. *persistent* (LSTM state carried over) on the second-episode SPL. A reduction under the persistent setting measures how tightly the stored representation is locked to the previous goal. For the foveated-fixed condition, all probes and behavioural runs use the ckpt.36 checkpoint ($\sim$174M frames, the last clean intermediate before the silent-corruption window, §4).

4 Results

4.1 Five conditions, same task

Table 1 collects the headline per-condition numbers this section returns to. All sighted conditions solve PointGoal at 96–99% success; blind reaches 93% at a longer training budget, reproducing Wijmans et al. [Wijmans et al., 2023]. Success is tightly bunched, so "the agents differ in what they do" is an unlikely explanation for the encoded-content gap we report (it does not strictly rule it out). The probe machinery itself is calibrated: distance-to-goal decodes robustly across all five conditions ($R^2 \in [0.81, 0.90]$; Appendix A for the companion plot), confirming that Ridge has the capacity to fit high-R^2 scalars when the hidden state contains them; and Hewitt–Liang label-permutation selectivity is high in the conditions where the GPS code itself is high (blind GPS / compass selectivity $+0.95/+0.81$; matched $+0.72/+0.56$), consistent with a genuine signal rather than label memorisation. Two training-stability caveats are disclosed in full in Appendix C: foveated-fixed uses ckpt.36 at $\sim$174M frames (last clean checkpoint before a silent NaN-gradient corruption at $\approx$182M); and matched-compute's 1×1 feature-map collapse, originally noticed as a training quirk, is what makes matched-compute a useful test of the encoder-bottleneck framing (we cannot from one condition alone fully separate the 1×1 feature-map effect from the lower input resolution; see §3.2 and §4.4).

Table 1: Per-condition headline. Probe R^2 columns are the 5-fold episode-level CV mean $\pm$ std on full-length deterministic-rollout trajectories (no per-episode step cap). *GPS* and *Compass* are world-frame. The bottleneck-vs-rich-encoder gap is in temporal *stability* of the code, not its mid-episode existence (Figure 2b): all five conditions decode top-layer GPS at $R^2 \in [0.6, 0.95]$ in the typical-episode window (steps 50–200); the rich-encoder long-tail collapse is what produces the sub-chance numbers below. Bold marks the H1-relevant ordering (blind $>$ matched $\gg$ rich-encoder).

Condition	Frames	SPL	Success	GPS R^2 (no-cap)	Compass R^2 (no-cap)
Blind	342M	0.59	0.93	$+\mathbf{0.95}\pm\mathbf{0.02}$	$+\mathbf{0.81}\pm\mathbf{0.08}$
Matched-compute	250M	0.67	0.98	$+\mathbf{0.78}\pm\mathbf{0.10}$	$+\mathbf{0.64}\pm\mathbf{0.10}$
Uniform	250M	**0.85**	**0.99**	-0.31 ± 0.86	$+0.36\pm0.23$
Foveated (fix)	174M	0.78	0.96	$+0.06\pm0.88$	$+0.07\pm0.69$
Fov-learned	250M	0.82	0.98	-2.43 ± 3.98	-1.34 ± 3.14

4.2 Encoder–memory race: a temporal-stability claim (H1)

Finding Visual-encoder capacity inversely tracks LSTM spatial encoding across our five conditions (Figure 2a). Blind agents encode current GPS in the recurrent state at $R^2 = 0.95 \pm 0.02$; matched-compute agents (48$\times$48 RGB input, 1×1 encoder feature map) at $R^2 = 0.78 \pm 0.10$; all three full-resolution-encoder conditions (uniform, foveated, foveated-learned) decode GPS at chance

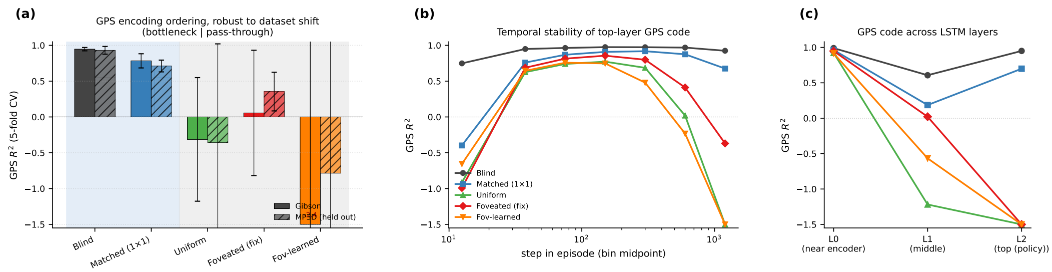


Figure 2: H1 evidence in three panels. *(a)* GPS R^2 (5-fold CV, Ridge $\alpha = 10$, full-length deterministic rollouts) on Gibson and held-out MP3D (300 episodes/condition; same checkpoints, no re-training). Blind and matched-compute (bottleneck conditions, blue background) decode world-frame GPS robustly on both datasets; uniform / foveated / foveated-learned (rich-encoder, grey background) do not, except for foveated recovering to a mild positive ($R^2 = +0.35$) on MP3D. The Compass and DtG numbers are in Table 1 (compass shows the same ordering at smaller magnitude; DtG decodes everywhere because every agent must track it; Appendix A for the companion bar plots). *(b)* Top-layer GPS R^2 binned by step-in-episode (out-of-fold prediction): all five conditions encode top-layer GPS in the typical-episode window (steps $\sim$ 25–200); the rich-encoder conditions then *overwrite* the code (negative R^2 in the long-tail $800+$ bin), while bottleneck conditions hold $R^2 > 0.7$ to step $800+$. The headline "rich-encoder $\approx$ chance" in panel (a) is therefore primarily long-tail behaviour. *(c)* Per-LSTM-layer GPS R^2 (single-fit on top-layer hidden states): all five conditions have GPS at Layer 0 (where the GPS sensor embedding enters the LSTM input vector — this lower-bound is at least partly a sensor-pass-through effect, not the network "computing" position); the divergence is at Layer 2, the policy readout.

($R^2 \in [-2.4, +0.06]$ with CV $\sigma \in [0.7, 4.0]$). Compass shows the same ordering at smaller effect size ($0.81 \to 0.64 \to 0.36 \to 0.07 \to -1.3$); DtG decodes robustly everywhere ($R^2 \in [0.81, 0.90]$). **[TODO: Whether this correlation reflects causation (encoder capacity *drives* LSTM encoding) is what the foveation strength sweep, log-polar control, and encoder feature-map probe in §4.4, and the encoder-resolution scaling sweep in Appendix E, test directly. Multi-seed runs of the existing five conditions are in progress to bound seed-to-seed variability.]**

Temporal stability across episode duration (H1's precise claim) The no-cap probe in Figure 2a does not separate "does the LSTM ever encode GPS" from "does the LSTM maintain that encoding as the episode lengthens." We test the temporal version directly with a step-binned out-of-fold probe (Figure 2b; OOF prediction so each bin's R^2 is from a probe never trained on that bin's episode). *All five conditions* encode top-layer GPS in the typical-episode window (steps 50–200, $R^2 = 0.6$–0.95); rich-encoder conditions are not at chance at this scale. They diverge from the bottleneck conditions in what happens *later*. Blind and matched-compute hold $R^2 > 0.7$ across all step bins out to $800+$. Uniform / foveated / foveated-learned peak at steps 100–200, then decay; by step 400–800 they are at or near chance, and in the long-tail $800+$ bin they are catastrophically negative (-0.37 to -5.67). The "rich-encoder $\approx$ chance" headline in Table 1 is therefore predominantly long-tail behaviour. *H1 is most precisely a claim about the temporal stability of the linearly-decodable top-layer GPS code, not its existence at any moment.* A separate lag-k probe corroborates: decoding GPS_{t-k} from $\mathbf{h}_t$ holds $R^2 = 0.92$ at $k = 20$ for blind and $R^2 > 0.5$ at $k = 20$ for matched, while rich-encoder conditions decode at chance for every $k \leq 10$.

Disambiguating Layer-0 sensor pass-through (H1's precise level of the network) A natural objection is that all five conditions have linearly-decodable GPS at LSTM Layer 0 ($R^2 > 0.91$, Figure 2c) — so how can H1 distinguish them? Layer 0 is where the GPS sensor embedding enters the LSTM input vector, so this lower bound is at least partly a sensor-pass-through effect rather than the network "computing" position. The interesting signal is the per-layer trajectory: bottleneck conditions *retain* the GPS code through Layers 1–2 to the top-layer policy readout, while rich-encoder conditions discard it (uniform Layer 2 GPS $R^2 = -1.56$, foveated -1.64, foveated-learned -10). Since the standard DD-PPO action head is a linear projection from the top-layer state $\mathbf{h}_2$, "linearly decodable at Layer 2" is precisely what the policy itself can use. *(Side observation, not yet explained: blind's Layer 1 GPS R^2 is lower (0.61) than its Layer 0 (0.99) or Layer 2 (0.95); the LSTM does not pass GPS through monotonically. Flagged as an artefact-or-mechanism question for follow-up.)*

Non-linear-probe disambiguation An MLP probe on the same top-layer state recovers position from rich-encoder networks (foveated $R^2 = 0.51$, foveated-learned $R^2 = 0.73$), *consistent with* position information persisting in a non-linear subspace — though we cannot from this alone exclude MLP exploitation of position-correlated features (wall distance, scene identifiers, episode-phase markers) rather than position itself. What this establishes is that the bottleneck-vs-pass-through divergence is not just about whether *any* probe can recover position; it is specifically about whether a *linear, top-layer* probe can. The encoder-resolution scaling sweep (Appendix E, in progress) and the foveation-strength sweep (§4.4) test the bottleneck framing more directly.

Robustness to dataset shift (MP3D) The H1 ordering is preserved on a held-out MP3D evaluation (300 episodes per condition, deterministic protocol; same trained checkpoints, no re-training; the MP3D bars in Figure 2a). Blind GPS R^2 moves from 0.95 (Gibson) to 0.93 (MP3D); matched moves from 0.78 to 0.71. Foveated recovers from 0.06 to a mild positive $+0.35$ on MP3D, and foveated-learned compass shows a large cross-dataset swing ($-1.34 \rightarrow +0.41$). We mark these single-seed cross-dataset shifts as observations requiring multi-seed replication before attributing them to a specific mechanism (Gibson-specific feature use by the foveated encoder is one candidate); the qualitative bottleneck-vs-pass-through partition is unaffected. Compass tells the same qualitative story across datasets (numbers in Appendix A).

Proposed mechanism We *propose* the following account, consistent with the data but not directly tested by a causal intervention. In PointNav, GPS and compass enter the LSTM input at Layer 0 directly, via a 32-d sensor embedding concatenated with the visual encoder's flattened output (Figure 2c shows GPS at Layer 0 across all five conditions, $R^2 > 0.91$). The encoder feature-map probe (§4.4, Figure 4) shows that none of the three sighted encoders linearly re-derive GPS from their visual stream: matched $R^2 = -3.14$, uniform -0.32, foveated -0.65. The encoder–memory race is therefore not about whether the encoder *re-derives* position; it is about how much *visual feature variety* the encoder hands to the LSTM, and whether that variety is enough to substitute for integrating the Layer-0 GPS-sensor signal across time. Bottleneck conditions (blind, no encoder; matched-compute, 1×1 spatial output $\rightarrow$ minimal feature variety per step) leave the LSTM with little visual structure to substitute, so it integrates the Layer-0 GPS signal across time and *maintains* a top-layer linearly-readable GPS code. Rich-encoder conditions (uniform / foveated / foveated-learned, 8×8 spatial output $\rightarrow$ many position-correlated visual features) give the LSTM enough visual variety to base actions on those features instead, and the top-layer GPS code drifts off as the episode progresses. Task success is tightly bunched, so a skill-gap artefact is unlikely to explain the encoding gap. The temporal dynamic in Figure 2b (rich-encoder GPS code emerging early at Layer 0 then decaying through Layers 1–2 to chance at the top layer) is the direct fingerprint of this account. **[TODO: A direct causal test of the mechanism — ablating the visual stream mid-rollout in rich-encoder agents to see whether the top-layer GPS code re-emerges, or perturbing the GPS sensor input mid-rollout in bottleneck agents to see whether SPL collapses — is left for future work.]**

Relation to Wijmans et al. Wijmans et al. [Wijmans et al., 2023] showed that blind PointNav agents develop spatial representations in LSTM memory. Our blind result ($R^2 = 0.95$) replicates this; matched-compute extends it to a condition that has visual input but a 1×1 spatial encoder output. The shared trigger across blind and matched is therefore the LSTM having minimal visual feature variety per step — not absence of vision per se, and not the encoder's ability to resolve position (no sighted encoder does, §4.4). §4.4 and Appendix E test the spatial-feature-variety reading more directly.

Bio parallel The pattern parallels the animal-navigation literature without depending on its mechanisms: congenitally blind humans develop enhanced hippocampal coupling and outperform sighted controls on non-visual spatial tasks [Kupers and Ptito, 2014]; rodent grid cells lose their hexagonal periodicity in darkness even with head-direction input intact [Chen et al., 2016], yet the animal still navigates — so spatial bookkeeping that the (vision-dependent) grid code can no longer carry has to come from somewhere else. We use these only as analogy: in both cases visual-pathway absence shifts spatial bookkeeping into a non-visual substrate, the same qualitative direction we measure in LSTM hidden states. We do not claim the agent and biological mechanisms are the same.

4.3 Format-level divergence across conditions (H2)

Lead with the behavioural test The most informative H2 measurement is also the most direct test of "do these representations matter?": *memory transplant* (Figure 3, right). For each donor–recipient pair, the donor drives the first t steps of an episode, its top-layer hidden state is copied into the

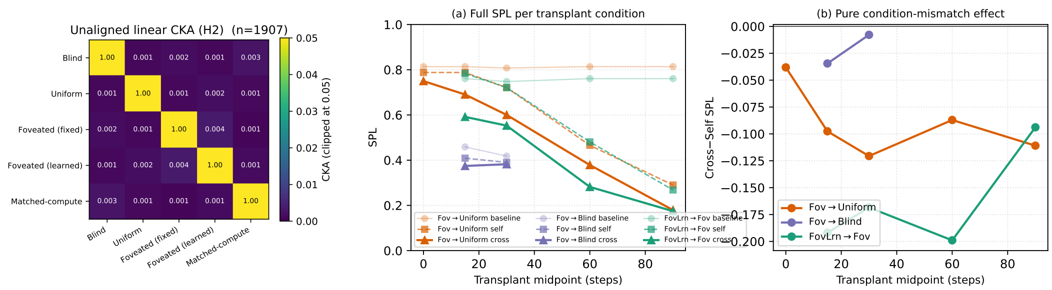


Figure 3: H2 convergent evidence. *Left:* Unaligned linear CKA [Kornblith et al., 2019] between top-layer LSTM hidden states (n=95,640 steps per condition under deterministic-rollout probes); every off-diagonal is below 10^{-4}. *Right:* Memory transplant across midpoints $\{0, 15, 30, 60, 90\}$ ($n = 100$–150 episodes/cell). *(a)* Baseline / self-transplant / cross-transplant SPL; at t=0 self- and cross- reduce to a protocol-noise floor. *(b)* Cross minus self (the isolated condition-mismatch component) is roughly stable for $t \geq 15$: foveated-learned→foveated incurs -0.17 SPL on average even though all pairwise CKA values sit at the noise floor. Probe-based similarity is uninformative at this scale; transplant reveals the functional gap.

Table 2: Probe-transfer GPS R^2 on deterministic-rollout hidden states: train on row condition, test on column. Self-accuracy diagonal is 0.89–0.99; every cross-condition transfer is catastrophically negative ($R^2 \ll -800$, often $\ll -10^4$), reflecting that a probe trained on one condition's hidden-state geometry does not just degrade on another — it predicts off-manifold values.

Train \ Test	Blind	Uniform	Fov-fix	Fov-lrn	Matched-compute
Blind	+**0.99**	−844	−11,301	−10,731	−55,706
Uniform	−1,383	+**0.90**	−7,046	−3,305	−110,317
Fov-fix	−2,091	−2,792	+**0.92**	−1,263	−39,100
Fov-lrn	−6,722	−2,423	−6,205	+**0.89**	−38,842
Matched-compute	−10,621	−4,832	−4,984	−58,169	+**0.96**

recipient at step t, and the recipient's policy drives the rest. The isolated condition-mismatch cost — cross-transplant SPL minus self-transplant SPL, which subtracts both a small t=0 protocol-noise floor and the rollout-divergence cost of mid-episode state replacement — is roughly stable across midpoints $t \geq 15$ and carries a clear ranking: foveated-learned→foveated incurs -0.17 SPL, foveated→uniform -0.10, and foveated→blind essentially neutral (-0.02). Pairs that look identically near-zero in linear-similarity measures differ markedly under transplant. Probe-based similarity is uninformative at this scale; the functional gap is what matters. **[TODO: The remaining cross-condition transplant pairs (e.g., blind→matched, matched→uniform) are not yet measured; running them would let us state the format-divergence finding as a complete 5×5 behavioural matrix rather than a representative subset.]**

Convergent linear-similarity evidence Three further linear-similarity measures converge on the same "different format" conclusion. *Pairwise CKA* [Kornblith et al., 2019] sits at the noise floor for every condition pair (Figure 3, left; off-diagonals all $< 10^{-4}$); within-condition split-half CKA is ≈ 1, so the near-zero off-diagonals are not estimator noise. *Cross-condition probe transfer* is catastrophically negative in every off-diagonal cell (Table 2, $R^2 \ll -800$, often $\ll -10^4$): a probe trained on one condition's hidden states does not just degrade on another, it predicts off-manifold values. *1-NN purity:* pooling 1500 random top-layer hidden states from each of the five conditions and asking whether each sample's nearest neighbour (Euclidean, full 512-d state) comes from the same condition, the answer is yes for $7500/7500$ in our sample (1.000 vs. chance 0.20). A t-SNE projection of the same pool (Appendix A) confirms visually that the five conditions occupy non-overlapping regions. **[TODO: Whether 1-NN purity holds at much larger sample size or under domain shift (MP3D-pooled, fov-shifted-pooled when available) is a further-investigate item.]**

Ruling out alternatives Self-probes succeed for every condition, so probe capacity is not the bottleneck. CKA is computed on a common-sample truncation, ruling out sample-size asymmetry. Task competence is tightly bunched (96–99% success across sighted conditions), so the divergence is not a skill difference. Transfer fails symmetrically (both directions), which rules out the possibility that one condition's code simply embeds another's. (CKA and probe transfer use common-sample truncation for fair cross-condition comparison; transplant uses full on-policy rollouts because the task is not truncatable. The two measurement scopes are deliberate, and their conclusions are consistent.)

Boundaries Our similarity tests are linear (CKA) or near-linear (t-SNE default); non-linear similarity measures could reveal higher-order shared structure. Our five sensors are diverse but not exhaustive; hybrid sensors (coarse uniform periphery + sharp fovea, depth-only inputs) might produce intermediate geometries.

4.4 Foveation: convergence on H1, divergence on H2 + behaviour

The five-condition results above place foveated-fixed and uniform in the same rich-encoder pass-through regime *on the headline H1 metric* (top-layer LSTM GPS R^2, Figure 2a): both decode at chance under no-cap probing, both peak in the typical-episode window then fade. *On every other axis we measure, however, foveated and uniform are not interchangeable*, summarised in Table 3. The cross-condition memory transplant from foveated to uniform incurs -0.10 SPL beyond the self-transplant noise floor (Figure 3); pairwise CKA between foveated and uniform sits at the numerical-noise floor ($< 10^{-4}$); 1-NN cluster purity on pooled hidden states is 1.000, so the two conditions occupy non-overlapping linear subspaces; the shortcut SPL drop is 21% for foveated vs. 41% for uniform; and on held-out MP3D, foveated GPS recovers to $R^2 = +0.35$ while uniform stays at -0.36 (Figure 2a, MP3D bars). The take-away: *within the rich-encoder regime defined by H1, foveation and uniform produce internally distinct representations whose differences manifest in transplant compatibility, dataset shift, behavioural memory reliance, and pairwise similarity — only the headline "current-state GPS R²" is the same*.

Table 3: Foveation vs. uniform at $\sigma_{\max} = 8$ Gaussian blur with the same ResNet-18 encoder. Same on H1, different on H2 / behavioural / generalisation. Sources: Table 1, Figure 3, Figure 5, Figure 4. Data are deterministic-rollout single-seed.

Measure	Foveated (fix)	Uniform	Verdict
LSTM GPS R^2 (Gibson)	$+0.06 \pm 0.88$	-0.31 ± 0.86	both chance, *indistinguishable*
Encoder $\rightarrow$ GPS R^2	-0.65 ± 0.28	-0.32 ± 0.08	both negative, overlapping
LSTM compass R^2 (Gibson)	$+0.07 \pm 0.69$	$+0.36 \pm 0.23$	uniform $\sim 5\times$ better
LSTM GPS R^2 (MP3D, held out)	$+0.35 \pm 0.27$	-0.36 ± 1.38	**differ by 0.71**
Shortcut SPL drop %	21%	41%	**uniform** $2\times$ **more brittle**
Memory-transplant cost (fov $\rightarrow$ uni)	-0.10 SPL beyond self-noise		**not interchangeable**
Pairwise CKA(fov, uni)	$< 10^{-4}$		**near-orthogonal subspaces**
1-NN purity (pooled)	1.000 vs. chance 0.20		**linearly separable**

These differences place foveated meaningfully apart from uniform along the H2 / dataset-shift / behavioural axes, even though their H1 signature is identical. The four in-flight experiments described next are the cleaner causal tests of *which* aspect of foveation drives these differences (sensor transform vs. gaze location vs. encoder bandwidth). Numbers from those will populate the corresponding sub-paragraphs in revision; current status is tracked in Appendix D.

Foveation strength sweep (F1–F4, in progress) We train four additional foveated agents at $\sigma_{\max} \in \{2, 4, 12, 20\}$ (alongside the existing $\sigma_{\max} = 8$), holding everything else fixed — gaze location, encoder stack, training budget, training pool. The expected signature: at low $\sigma_{\max}$ the foveated agent should look like uniform (encoder still resolves world-frame); at high $\sigma_{\max}$ it should approach matched-compute (encoder cannot resolve world-frame, LSTM compensates). The R^2 vs. $\sigma_{\max}$ curve directly tests the encoder–memory race as a continuous lever, not a binary one.

Spatial sampling vs. blur (F3 log-polar control, in training) Gaussian blur removes high-frequency content but preserves *spatial layout*: a $\sigma_{\max} = 8$ foveated image still feeds the encoder 256×256 pixels, so the ResNet-18 stack still produces an 8×8 spatial feature map. A log-polar foveation [Strasburger et al., 2011, Rosenholtz, 2016] resamples the image with a non-uniform retinal grid (e.g. 64×64): the encoder's spatial output drops to $\sim 2\times2$, much closer to matched-compute's 1×1 than to uniform's 8×8. This is the architectural commitment "foveated vision" makes in biology, and it isolates the variable our sharpened H1 mechanism (§4.4, encoder feature-map paragraph) identifies as the trigger: *spatial-feature variety in the encoder output*.

Predictions, falsifiable. If the encoder–memory race is driven by encoder spatial output dimensionality (not by "bottleneck of any kind"), then log-polar should produce an LSTM GPS R^2 *between* matched-compute (current $+0.78$) and uniform (current ≈ 0), likely ≥ 0.3, with corresponding behavioural shifts (longer paths than uniform, larger shortcut SPL drop closer to blind / matched). If instead log-polar matches uniform's near-chance H1 readout, the mechanism is not pure encoder-output-dimensionality and we need to reframe (e.g. allow the encoder to learn task-useful features from any spatial layout, rendering "bandwidth" the wrong axis). Either outcome is informative; F3 is in training (~14M / 250M frames at submission, ETA 2–3 days). Numbers will populate this paragraph in revision; the prediction was written before the result was available.

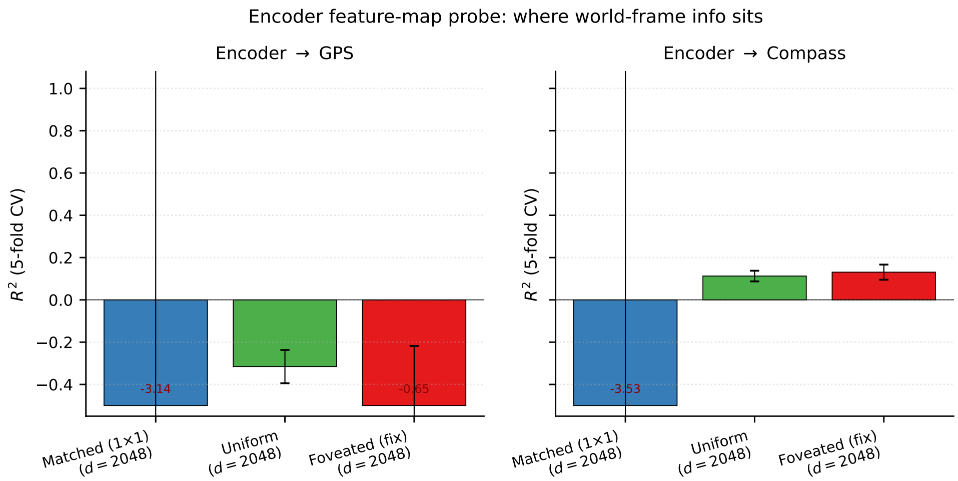


Figure 4: Encoder feature-map probe (post ResNet-18, pre-LSTM, 5-fold CV Ridge). Encoder dim $d = 2048$ (compressed flatten) for all three sighted conditions. None of the three encoders linearly decode GPS (matched $R^2 = -3.14$, uniform -0.32, foveated -0.65); the encoder–memory race is therefore not about encoder-level GPS information availability. Compass shows a small positive on uniform / foveated ($+0.11, +0.13$); matched compass is also chance.

Where is position information held in the foveated agents? (encoder feature-map probe) Our H1 evidence comes from probing the LSTM hidden state. The complementary question — can the encoder's output already resolve position, or is the LSTM the only place world-frame structure lives? — we settle by probing the encoder feature map directly (post ResNet-18, pre-LSTM). We forward-hook the encoder's compression layer (a final 1×1 conv that flattens spatial features into a 2048-d vector before they enter the LSTM input) and run the same Ridge probe. The result is striking: *none of the three sighted encoders linearly decode GPS* (matched $R^2 = -3.14 \pm 5.91$, uniform -0.32 ± 0.08, foveated -0.65 ± 0.28; Figure 4). Compass is at chance for matched (-3.53) and slightly positive but well below ceiling for uniform / foveated ($+0.11, +0.13$). The encoder–memory race is therefore not about encoder-level information availability — all three encoders effectively discard linear position info. *This sharpens the H1 mechanism*: the divergence happens inside the LSTM (or at the GPS-sensor pass-through that enters Layer 0), not at the visual encoder. Bottleneck conditions integrate the GPS-sensor input across time because the visual stream gives the LSTM minimal feature variety to substitute for it; rich-encoder conditions have richer (though still position-blind) visual features that the LSTM uses instead, overwriting the Layer-0 GPS code as the episode progresses. Foveated specifically does not differ from uniform on this measure (-0.65 vs. -0.32, overlapping uncertainty), so its pass-through behaviour is consistent with "rich-encoder" rather than a foveation-specific effect at the encoder level.

Foveated-shifted (gaze location, in progress) Gaze location is treated separately as H3 (§4.5), but the foveated-shifted control directly informs the foveation slot too: training a foveated agent with hardcoded static gaze at $(0.49, 0.62)$ matches foveated-learned's collapsed gaze location while removing the learned-gaze module from the optimisation path. If foveated-shifted does *not* reproduce foveated-learned's specific behaviours (Gibson–MP3D compass swing, transplant ranking), those behaviours can be attributed to the optimisation-path interaction and not to the static gaze location alone.

Why this slot exists Our title and framing focus on foveated vision; the present five conditions place foveated-fixed and foveated-learned within a framework but do not isolate foveation-specific effects (under simple Gaussian blur both behave like uniform). The four experiments above target specific foveation-specific predictions of the encoder–memory race framework. Each is a falsifiable test, designed and submitted at the time of writing. None depends on the others to be informative.

What we can already conclude about foveation From the existing data alone (Table 3), foveation under $\sigma_{\max} = 8$ Gaussian blur is *not a copy of uniform*. It converges with uniform on H1 but diverges from it on H2 (subspace orthogonality, 1-NN separability), behavioural memory reliance (shortcut drop 21% vs. 41%), transplant compatibility (-0.10 SPL gap), and dataset-shift recovery ($+0.71$ R^2 difference on MP3D). The encoder feature-map probe additionally shows that the encoder discards linear position info regardless (-0.65 for foveated, -0.32 for uniform; Figure 4), so the differences above cannot be attributed to encoder-level GPS preservation: *whatever foveation does differently from uniform, it does it inside the LSTM, in subspaces that the linear-GPS probe cannot see*. The in-flight experiments below probe *what* that "something different" is by varying foveation strength, sensor transform (Gaussian-blur vs. log-polar spatial sampling), and gaze location.

4.5 Gaze location as a second axis (H3)

H1 and H2 identify sensor structure as one axis on representational content. H3 asks whether gaze *location* — where the foveated window is centred — is a second, dissociable axis. We test the two forms described in §3.3 and previewed in the Introduction: an active-gaze form (the minimal learned-gaze module should move its gaze to task-useful locations), and a gaze-location form (two foveated agents whose gaze is centred at *different static* locations should produce different content). The active-gaze form is out of scope of our minimal gaze decoder and we do not claim anything about it; the gaze-location form is what our controlled experiments test.

The learned-gaze module collapses Our minimal learned-gaze implementation (deterministic sigmoid MLP, slow-gaze updates once per 128-step rollout segment; Appendix B) does not produce active gaze: after 250M frames the gaze output is frozen at $(0.49, 0.62)$ with per-dimension std $\leq 2.5 \times 10^{-4}$ (numerical noise). Two anti-collapse regularisers (Appendix B) restore gaze variance only at the cost of task performance. We explicitly do not claim this shows active gaze cannot be learned — our gaze decoder is minimal by design (to be a controlled variable, not a model of active vision), and richer architectures (stochastic gaze, information-seeking intrinsic reward, attention-based token selection) would need separate investigation. What the collapse gives us for free is an opportunity to test the gaze-location form: after training, we have an agent that is *effectively foveated with a static gaze* 30 *pixels below image centre*.

Gaze-location test (foveated-shifted, in progress) Both rich-encoder foveated variants sit in the pass-through regime under current-state probing (Table 1), so the H3 test is whether *within* the pass-through regime, gaze location shifts content along other axes — non-spatial feature usage, behavioural memory-reliance (shortcut, Figure 5), or transplant compatibility. To isolate gaze location from training-dynamics confounds, we train a foveated-shifted control: an agent identical to foveated-fixed except gaze is hardcoded at $(0.49, 0.62)$ instead of $(0.5, 0.5)$ — matching foveated-learned's collapsed gaze position but with no learned-gaze module in the optimisation path. **[TODO: Training in progress at submission; results will populate this paragraph.]**

Shortcut discovery (behavioural memory-reliance) We separately measure how tightly each policy couples action choice to integrated recurrent memory: run paired episodes in the same scene with fresh goals, compare second-episode SPL under *reset* vs. *persistent* memory. Larger relative SPL drop = more brittle memory coupling: blind loses 52%, uniform 41%, foveated 21%, matched 18%, foveated-learned 15%. Plotting these against the H1 GPS R^2 (Figure 5) reveals a 2×2 dissociation: blind sits in the "has linear GPS, uses memory" quadrant (consistent with H1); foveated-fixed and foveated-learned sit in "no linear GPS, low memory reliance" (also consistent). The two off-diagonal cells are the interesting ones. *Matched-compute* "has GPS code but ignores memory" (high probe R^2, low shortcut drop) — it has a linear top-layer cognitive map but weights it less in the policy output. *Uniform* "has no linear GPS yet relies heavily on memory" (low probe R^2, high shortcut drop) — it integrates memory across episodes in a form that is not linearly readable as world-

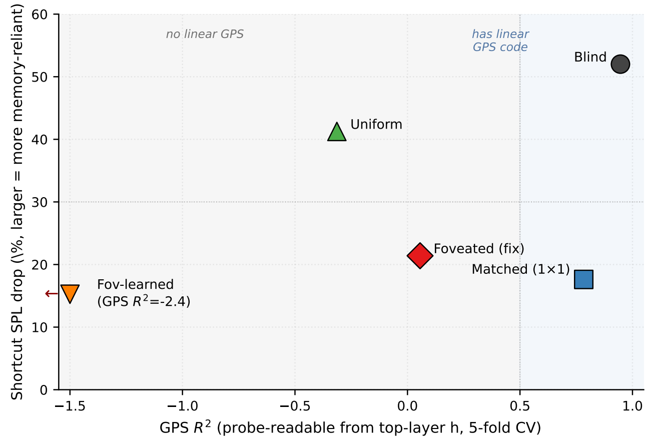


Figure 5: Probe-readable GPS (H1) vs. behavioural memory reliance (shortcut). The shortcut experiment runs paired episodes in the same scene with fresh goals and compares second-episode SPL under *reset* (LSTM zeroed) vs. *persistent* memory; larger drop = more memory-reliant policy. *x-axis:* GPS R^2 (5-fold CV mean) from Figure 2a. *y-axis:* relative shortcut SPL drop (%). 20 Gibson scenes $\times$ 10 paired episodes per condition. Fov-learned's $R^2 = -2.43$ is clipped at -1.5 (arrow). The 5 conditions populate a 2×2 dissociation: blind "has GPS code, uses memory"; foveated-fixed and foveated-learned "no linear GPS, ignores memory"; **matched** and **uniform** are the two opposite anomalies (matched: "has GPS but ignores memory"; uniform: "no linear GPS yet still memory-reliant").

frame position, plausibly visual-landmark / scene-history features that condition the policy without recovering position. The take-away: "decodable from h" and "policy uses for decision-making" are distinct constructs — a probe shows what is *readable*, not what the policy actually *relies on*, and the two can dissociate in either direction.

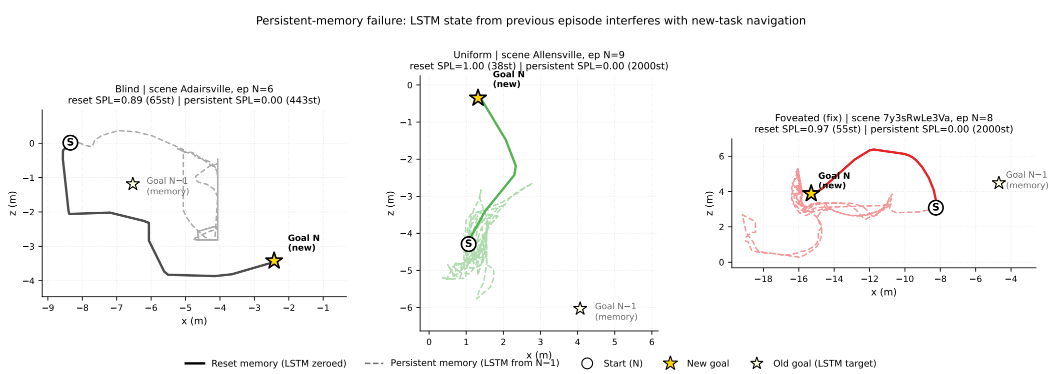


Figure 6: Persistent-memory failure made visual: trajectories from one paired-episode run per condition where reset memory succeeds and persistent memory fails. Same scene, same start position, same goal between the two memory conditions; the only difference is whether the LSTM is zeroed (reset) or carries over from the previous episode (persistent). Blind (left) shows the cleanest interpretation of the shortcut SPL drop: persistent agent oscillates around the *old* goal location (light star, $N-1$) instead of going to the new one (gold star, N), consistent with the "LSTM still encodes previous task" reading. Uniform (centre) and foveated (right) persistent runs wander more diffusely without obviously locking onto the old goal — consistent with their hidden state encoding scene-history / visual features rather than a position-readable memory of the previous task. The contrast in trajectory length is also informative (blind reset 65 steps vs. persistent 443; uniform 38 vs. 2000 max; foveated 55 vs. 2000).

Figure 6 shows three example paired episodes (one per condition, manually selected from cases where reset SPL > 0.5 and persistent SPL < 0.2 with persistent steps ≥ 30, so the failure mode is visible rather than an immediate STOP). Blind's persistent trajectory clusters around the previous-episode goal location, supporting the literal "locked onto old goal" reading; uniform and foveated persistent trajectories wander more diffusely. The visualisation cannot distinguish "locked onto old position" from "locked onto old visual context" but the failure modes look qualitatively different across the two regimes, consistent with the 2×2 dissociation in Figure 5. **[TODO: Both anomalies rest on single-seed runs of both the probe and the shortcut measurement; multi-seed replication is in progress and is what would let us promote the 2×2 dissociation from a candidate dissociation to a robust finding. A direct measurement of "policy GPS reliance" (e.g., ablating the GPS sensor input mid-rollout and measuring the SPL drop) would complement the shortcut signal in distinguishing "policy reads the GPS code" from "policy reads non-GPS memory" — the matched-compute anomaly predicts a small SPL drop, the uniform anomaly predicts a large one.]**

4.6 Boundaries and additional probes

Occupancy controls Two probes fail to discriminate between conditions and so define the boundary of our claims. Coarse occupancy ("have I visited this region?") decodes above chance in every condition, so one-bit spatial working memory is ubiquitous and not part of the H1–H3 story. Metric 20×20 occupancy reconstruction fails everywhere, consistent with the SPACE benchmark's finding that fine-grained layout requires non-linear probes or longer horizons [Ramakrishnan et al., 2025]: our claims are about representational *format* and *content*, not about high-resolution metric layout.

Goal vector No condition stores the ego-frame goal vector linearly: goal *distance* (DtG) decodes strongly across conditions ($R^2 \in [0.81, 0.90]$, validating the probe machinery), while goal *direction* and the full 2-D ego-to-goal vector are at chance or negative for every condition (Appendix A, Figure 9). This is unsurprising — PointGoal provides the goal vector as a per-step input, so the policy has no incentive to re-encode it — and is not part of the H1/H2/H3 story.

Per-unit spatial information and population structure A population coding analysis (Appendix A, Figure 8) sharpens the H1 picture in a direction we did not expect. Per-unit spatial information [Banino et al., 2018]: rich-encoder conditions have a few units with *higher* peak spatial information (≈ 2 bits) than blind (≈ 1 bit max). What separates conditions is what those units actually encode — a sparse-decoding curve (probe GPS using only the top-k spatial-info units) reveals that blind reaches $R^2 \approx 1.0$ at $k = 512$, matched climbs to $R^2 \approx 0.5$, and the three rich-encoder networks fail to decode GPS at any k. The peaked units in rich-encoder conditions therefore appear to encode features correlated with position (visual landmarks, trajectory-phase markers) rather than position itself, while blind's flatter, lower-peak distribution is what actually carries the linear position code. Intrinsic dimension via PCA-95% confirms this: blind's representation is the most compressed (180 dimensions), matched intermediate (271), and rich-encoder networks the highest (291–303). *Caveat:* the spatial-info computation depends on the per-scene grid resolution and we do not claim a sharp dissociation between "population code" and "few dedicated cells"; the qualitative direction (peak per-unit info $\neq$ population GPS readout) is what we report.

5 Discussion

5.1 Two orthogonal axes: sensor structure and gaze location

Our findings have a common interpretation that we call the *encoder–memory race* (we present it as a candidate unifying account, not a proven mechanism; a direct causal test is left for follow-up). Whether the LSTM *maintains* a linearly-decodable world-frame spatial code at the top layer depends on whether the visual encoder can resolve that information from the current frame. When the encoder cannot — blind (no encoder) or matched-compute (1×1 feature map) — the LSTM keeps the GPS code stable across episode duration (H1, §4.2; Figure 2b). When the encoder can — uniform, foveated, foveated-learned all use full-resolution ResNet-18 encoders — the LSTM still encodes top-layer GPS in mid-episode (steps 50–200), but *overwrites* the code as visual-encoder features accumulate, leaving long-tail steps at chance or below. The encoder–memory race is therefore best read as a temporal-stability claim, not an existence claim. The Layer-0 finding (Figure 2c) is consistent: all conditions have GPS at Layer 0 (where the sensor embedding enters); the divergence is at Layers 1–2 *and* as the episode progresses. H2 (§4.3) cuts orthogonally: *what* the LSTM encodes is in different linear subspaces across conditions even where current-state position decodes at chance, and that linear difference matters for downstream behaviour (cross-condition memory transplants drop SPL more than transplant mechanics alone). H3 (§4.5) asks whether, *within* the rich-encoder pass-through regime, gaze location is a second representational-content axis; the clean foveated-shifted causal control that answers this is in training at submission time. The same encoder-capacity axis that collapses the original "foveated vs. uniform" question (both sit in the pass-through regime) surfaces a clean scaling prediction: GPS decodability should fall monotonically with encoder resolution, which we begin to test in Appendix E.

5.2 Why this, not alternative accounts

We rule out four alternative explanations.

Task-competence differences. Every condition solves PointGoal at 93–99% success, so the encoded-content gap cannot be attributed to differences in navigation skill (Table 1).

Probe-capacity / probe-fitting artefacts. DtG decodes robustly across all five conditions ($R^2 \geq 0.81$), demonstrating that Ridge has the capacity to fit high-R^2 scalars when the hidden state contains them. Where GPS does decode (blind, matched), Hewitt–Liang label-permutation selectivity is correspondingly high; where GPS is at chance, selectivity is also at chance because real and control probes face the same near-zero target-information ceiling. The chance-level GPS/compass R^2 for rich-encoder conditions is therefore a property of the hidden state, not of the probe.

Sample-size / trajectory-distribution artefacts. CV σ spans come from the same 500-episode pool with ~ 100–400-step rollouts under identical deterministic evaluation; the protocol confound that haunted an earlier draft (4-step quasi-static trajectories under stochastic sampling) is documented and retired in §3.3.

Linear-probe artefacts. Two complementary checks address this. *(a) Behavioural:* the memory-transplant intervention (§4.3) goes outside the probe framework and reproduces the bottleneck-vs-pass-through ordering, with cross-condition transplants dropping SPL by 0.19–0.21 between rich-encoder pairs. *(b) Non-linear probe:* a 2-layer MLP on the same top-layer hidden states recovers $R^2 \in [0.51, 0.73]$ from the rich-encoder conditions where Ridge sees chance. We interpret this as consistent with position information persisting in a non-linear subspace at the top layer, but emphasise the limits: an MLP probe can also exploit features merely *correlated* with position (wall distance, scene identifiers, episode-phase markers), so this number alone cannot disentangle "non-linear position code" from "MLP regressing through position-correlated features." What it does establish is that the bottleneck-vs-pass-through divergence is not just about whether *any* probe can recover position — it is specifically about whether a *linear* top-layer probe can. Since the standard DD-PPO action head is a linear projection from $\mathbf{h}_2$, the linear-probe metric is precisely what the policy itself can use — so the H1 distinction maps directly onto policy-accessible encoding regardless of how the MLP-probe interpretation resolves. The encoder-resolution scaling sweep (Appendix E, in progress) tests the bottleneck framing more directly by varying input resolution while holding everything else fixed.

5.3 Biological precedent as convergent evidence

Each of our three findings has a parallel in the animal-navigation literature, which we take as one piece of convergent (not conclusive) evidence that the effects are not specific to a single LSTM architecture. Compensatory memory under reduced sensory access is documented in congenitally blind humans, who develop enhanced hippocampal coupling and outperform sighted controls on non-visual spatial-working-memory tasks [Kupers and Ptito, 2014], and in rodents whose grid-cell periodicity collapses in darkness despite intact head-direction inputs [Chen et al., 2016]—both format-level, not just strength-level, effects. Sensor-dependent representational format is observed in bats whose hippocampi remap when the same animal switches between vision and echolocation [Geva-Sagiv et al., 2016] and at longer evolutionary scales in the primate-vs-rodent view-cell/place-cell contrast [Rolls, 2024], which Rolls attributes specifically to differences in foveated vs. near-panoramic sampling statistics. Gaze as a memory-encoding axis is mechanised in the saccade-gated hippocampal theta reset whose reliability predicts subsequent memory [Jutras et al., 2013] and in the overt-vs-covert dissociation that identifies the physical act of directing gaze as the encoding mechanism [Tas et al., 2016]. We do not claim our agents implement these biological mechanisms, nor that the parallel structure settles the generality question; we do claim that the parallels make a pure single-architecture artefact less likely, and that the controlled agent setting complements biological comparisons where sensor type, architecture, and training objective cannot be independently manipulated.

5.4 Implications for ML practice and architectural scope

Within the recurrent PointNav setting we study, three implications follow. (i) *Encoder capacity correlates inversely with linear, top-layer, policy-accessible LSTM spatial encoding* across our five conditions. If the correlation reflects causation in this setting — which the encoder-resolution scaling sweep (Appendix E, in progress) tests directly — it would also be a practical training-time lever for inducing cognitive-map-like internal representations: impose an encoder bottleneck. We do not yet claim this generalises beyond PointNav with our specific architecture and training regime. (ii) *"More pixels" is not the right lens within this setting:* the linear top-layer GPS code is non-monotonic in raw pixel count — uniform 256×256 (full encoder, 8×8 feature map) lacks it; 48×48 with the same encoder stack (matched-compute, 1×1 feature map) has a strong one. The encoder feature-map probe (§4.4) shows neither encoder linearly preserves GPS, so the mechanism is not encoder-level position retention; we attribute the pattern to the encoder's spatial-feature variety per step (i.e. how many position-correlated visual features the encoder hands the LSTM), with the scaling sweep needed to confirm and the log-polar control to test the prediction directly. (iii) *Attention policy (H3, pending):* within the rich-encoder regime, we predict that shifting the static gaze location alters what the LSTM encodes; the foveated-shifted causal control is in training at submission time and will test this directly (§4.5).

Beyond the recurrent setting (untested) We use a Wijmans-style recurrent (LSTM) backbone for direct comparability with the emergent-map literature. The three implications above are established only in this setting, but we speculate about where they plausibly extend. Transformer-based embodied agents build their internal state by accumulating information via learned attention, which is structurally similar to our gaze location in the sense that it selects which observation content is read into the representation; our H3 finding therefore suggests a testable prediction for transformer navigators—strong constraints on attended tokens should produce content-level dissociations of the sort we observe—but the prediction has not been verified and is left for future work. The sensor-structure axis (H1, H2) is a claim specifically about RL recurrent navigation agents trained under partial observability; whether equivalent representational-content effects arise in supervised visual learners, non-visual modalities, or multimodal systems is an open empirical question that the present experiments do not resolve.

5.5 Limitations

(i) *Intermediate checkpoint for foveated-fixed:* all foveated-fixed analyses use ckpt.36 at $\sim$174M, the last checkpoint before the silent-corruption window (Appendix C); the other four conditions use the final converged checkpoint. (ii) *Minimal learned-gaze architecture.* Our learned-gaze module is a deterministic sigmoid MLP with slow-gaze rollout updates, chosen for computational compatibility with DD-PPO rollouts and simplicity as a controlled variable, not as a competitive active-gaze model. Its collapse under pure task supervision is specific to this architecture (and so far to a single seed; multi-seed replication is in progress); richer gaze mechanisms (stochastic gaze with entropy bonus, information-seeking intrinsic reward, attention-based token selection) would need separate investigation. We therefore do not claim that implicit task pressure cannot produce active gaze in general; we claim only that our minimal decoder collapsed, and we use the collapsed-gaze position as an empirical starting point for the gaze-location test (foveated-shifted control, §4.5) — a test whose logic does not depend on the gaze decoder working. (iii) *Cross-heading probe:* the proposal included a cross-heading generalisation probe (train on heading A, test on opposite heading). Our implementation returned "insufficient samples" sentinels at our probe sizes and does not appear in our results. (iv) *Foveation model:* Gaussian blur with quadratic falloff is a simplification; log-polar or multi-scale pyramid models may behave differently. A log-polar control (F3) and a $\sigma_{\max} \in \{2, 4, 12, 20\}$ strength sweep (F1–F4) are in training at submission and will populate §4.4 and Appendix D. (v) *Linear probes only:* non-linear probes might reveal structure Ridge regression misses, particularly for occupancy maps. (vi) *Architecture scope:* we use a recurrent (LSTM) backbone to remain comparable with the emergent-map literature we build on. The core claims (sensor structure and gaze location as representational-content axes) should hold for transformer-based navigation agents whose internal state is built by learned attention, but directly verifying this requires re-running the five-condition ablation on a transformer backbone and is left for future work; see §5 for the specific predictions.

6 Conclusion

We ran a controlled five-condition ablation isolating the effect of visual-input structure on the recurrent representations learned by an otherwise-identical DD-PPO navigation agent. Three findings emerge.

(1) Encoder–memory race. LSTM top-layer world-frame GPS encoding is *temporally stable* when the visual encoder cannot resolve world-frame structure on its own, and *transient* when it can. All five conditions encode top-layer GPS in the typical-episode window (steps 50–200, $R^2 \in [0.6, 0.95]$). Blind and matched-compute then maintain that code across episode duration (blind $R^2 = 0.92$ at step $800+$; matched $R^2 = 0.68$); uniform / foveated / foveated-learned overwrite it (long-tail R^2 at chance or below). The headline "rich-encoder ≈ chance" GPS R^2 in Table 1 is therefore primarily a long-tail statistic, not a mid-episode property. Per-layer probing additionally shows all five conditions have linearly-decodable GPS at Layer 0 (where the sensor embedding enters), and an MLP probe at Layer 2 still recovers position from rich-encoder conditions ($R^2 \in [0.51, 0.73]$); we cannot from MLP recovery alone exclude exploitation of position-correlated features. H1 is most precisely a claim about what a *linear, top-layer, late-episode* probe can read — which is also what the standard DD-PPO action head reads.

(2) Condition-specific representational format. Even where rich-encoder LSTMs are spatially pass-through, their hidden-state subspaces are linearly separable to single-nearest-neighbour granularity: pairwise CKA at the noise floor ($< 10^{-4}$), cross-condition probe transfer catastrophically negative ($R^2 \ll -800$), 1-NN purity 1.000 across pooled samples, and cross-condition memory transplants that drop SPL by 0.19–0.21 between rich-encoder pairs (isolation metric up to -0.17). Whatever non-spatial task-relevant features each LSTM carries, it carries them in its own linear subspace — so much so that a probe trained on one condition predicts off-manifold values on another.

(3) Gaze-location test (pending) and a having–vs–using dissociation. Whether gaze location is a second representational-content axis *within* the rich-encoder regime is what H3 asks; the cleanest test (foveated-shifted training with hardcoded gaze at the learned-gaze module's collapsed location) is in progress at submission and will report in §4.5. Independently of that pending test, plotting probe-readable GPS (H1) against behavioural memory reliance (shortcut SPL drop) reveals a 2×2 dissociation that the H1 ranking alone hides: matched-compute has a linear top-layer GPS code but its policy weights it less than the probe predicts; uniform has no linear GPS yet still relies heavily on memory, plausibly via non-spatial scene-history features. The two opposite anomalies suggest "probe-readable" and "policy-used" are distinct constructs in this setting; both rest on single-seed runs and we mark them as candidate dissociations pending multi-seed replication.

The pattern parallels three threads of animal-navigation research — enhanced hippocampal coupling in congenitally blind humans, the disruption of rodent grid-cell firing in darkness with navigation nonetheless preserved, and cross-modality hippocampal remapping in bats. We use these as analogy only: the convergent direction across systems is suggestive of a sensor–memory coupling principle, but our experiments do not test architectures other than LSTM, so we do not claim the principle is architecture-independent. Open questions: (i) whether the encoder-bottleneck ordering extends to finer-grained resolution sweeps (Appendix E); (ii) whether H3 stands on the clean causal test; (iii) whether transformer-based navigation agents reproduce the encoder–memory race; (iv) whether equivalent effects shape representations in non-navigation embodied tasks; (v) multi-seed replication of all five conditions to bound seed-to-seed variability of the H1 numbers.

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6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer:
answerYes

Justification: §3.1 specifies training pool and eval set; §3.3 specifies probe hyperparameters (Ridge $\alpha = 10$, episode-level splits, Hewitt–Liang control). Appendix A gives the ablation's loss formulation and coefficient.

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Justification: We report single-seed numbers per condition. This is documented as Limitation (ii) in §5.4. Multi-seed replication for tighter confidence intervals is in the immediate follow-up.

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8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer:
   answerYes

   Justification: §3.7 reports per-condition training budget (5–7 days on 2×V100 for 250M frames) and probing budget (<1h per condition on one V100).

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   Answer:
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10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

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    answerYes

    Justification: §5.3 discusses implications — sensor design as a tunable axis for interpretable embodied agents. Negative societal impacts are minimal: the work is fundamental research on simulated indoor navigation with no direct deployment pathway.

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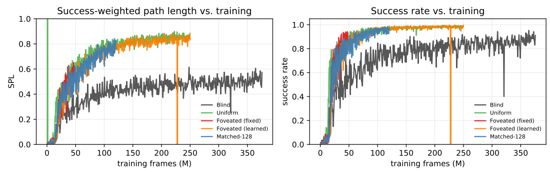
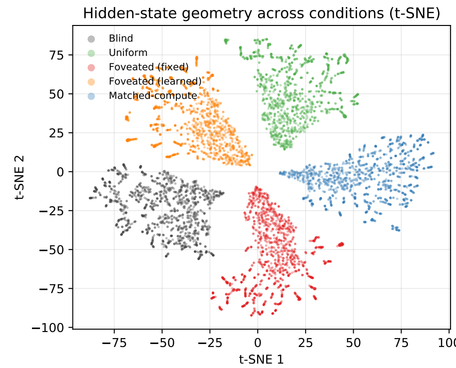
A Supplementary visualisations



(a) Training curves (TensorBoard event-file scalars). All four sighted conditions converge to SPL 0.82–0.85 and success $> 97\%$ by $\sim$100M frames; blind converges to SPL $\sim$0.50 over a longer horizon. The glitch at frame $\approx$ 228M (fov-learned) is a resume-boundary artefact in the scalar log.



(b) t-SNE projection of top-layer LSTM hidden states (deterministic-rollout probes), pooled across all five conditions (perplexity 30). The five conditions occupy non-overlapping regions of the 2-D embedding — a visual confirmation of the near-zero CKA result (§4.3).

Figure 7: Supplementary visualisations referenced from the main text.

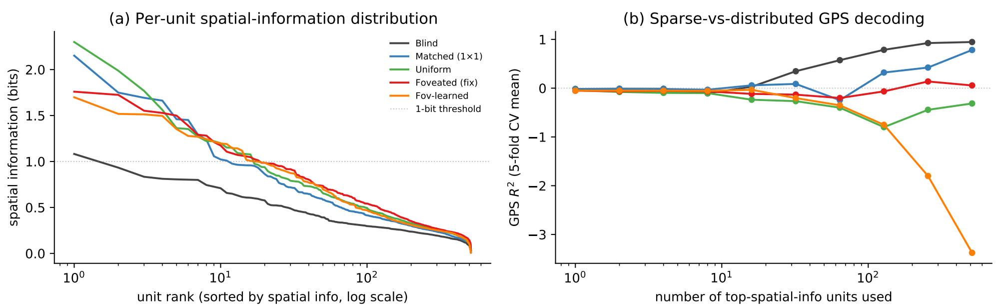


Figure 8: Population coding analysis (referenced from §4.6). *(a)* Per-unit spatial-information distribution: each curve is one condition's units sorted by spatial information (descending), x-axis log scale. Counter-intuitively, rich-encoder conditions (uniform / foveated / fov-learned) have a few more peaked spatially-tuned units (max ≈ 2 bits) than bottleneck conditions; blind has the flattest distribution (max ≈ 1 bit, no unit reaches the dashed 1-bit threshold). *(b)* Sparse-vs-distributed GPS decoding: probe R^2 when only the top-k spatial-info units are used as features. Blind reaches $R^2 \approx 1.0$ at $k = 512$; matched climbs to $R^2 \approx 0.5$; rich-encoder networks fail to decode GPS at any k even though their top units carry higher per-unit spatial information. The combined picture: *higher per-unit spatial information $\neq$ position readout*. Rich-encoder peaked units appear to encode features correlated with position (e.g. visual landmarks, trajectory phase) rather than position itself.

B Gaze-Diversity Auxiliary Loss (H3 Ablation)

Motivation In the first foveated-learned training run the gaze decoder collapsed (per-dimension std $\approx 5 \times 10^{-4}$). We attribute this to two mechanisms: (i) once the sigmoid output saturates, its derivative $\sigma'(x) \to 0$ attenuates gradient signal; (ii) consistent-gaze views yield more predictable visual features, so the reward-gradient path has no explicit incentive for variety.

Target-variance hinge We add $\mathcal{L}_{\text{div}} = \lambda \cdot \text{relu}(\sigma^2_{\text{target}} - \text{Var}_{\text{batch}}[g])$ where $g \in [0, 1]^2$, $\text{Var}_{\text{batch}}$ averaged over the two gaze dimensions, $\sigma^2_{\text{target}} = 0.01$ (std ≈ 0.1), $\lambda = 0.01$. The hinge disappears once variance exceeds the target. We also evaluated an un-capped $-\lambda \cdot \text{Var}[g]$ variant as a negative control.

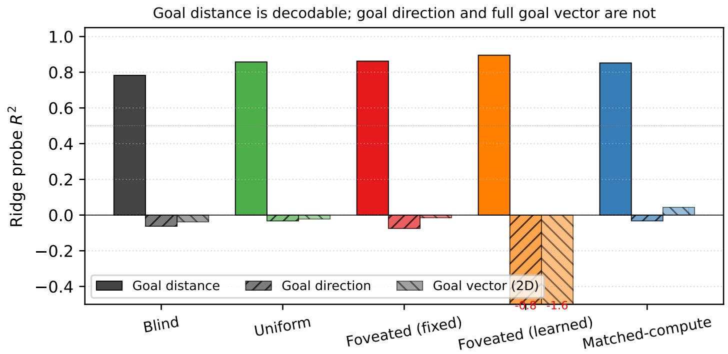
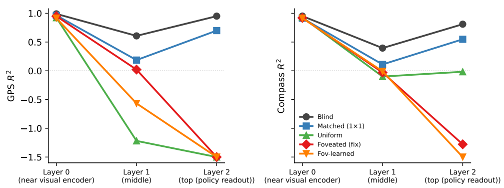
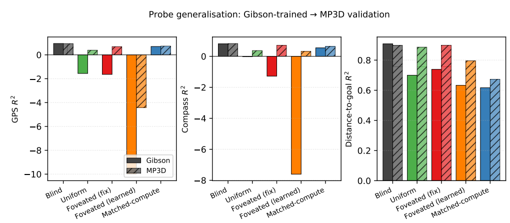


Figure 9: Goal-vector probe R^2 (referenced from §4.6). Goal *distance* (DtG) decodes strongly across all conditions ($R^2 \in [0.81, 0.90]$, validating probe machinery); goal *direction* and the full ego-frame goal vector are at chance or negative everywhere. PointGoal provides the goal vector as a per-step input, so the policy has no incentive to re-encode it.



(a) Per-LSTM-layer GPS / compass probe R^2 across all three layers, all five conditions (single train/test fit; the 5-fold CV mean reported in Table 1 matches qualitatively but with different magnitudes for rich-encoder conditions). Compass tells the same story as GPS at smaller magnitude.



(b) Companion to Figure 2d (which uses 5-fold CV mean): single train/test fit on the same checkpoints; the qualitative bottleneck-vs-pass-through partition is the same.

Figure 10: Per-layer and MP3D companion figures (referenced from §4.2).

Pilot 1 (uncapped, job 2840256) After 10 M frames, gaze variance exceeded that of a uniform distribution over $[0, 1]^2$ (std $[0.43, 0.42]$ vs. theoretical max 0.289): gaze was essentially random. Task performance collapsed: SPL plateaued at 0.05, running-mean metrics transitioned to NaN at ~ 8 M frames. The uncapped regulariser destroyed the consistent-foveation signal the policy needs.

Pilot 2 (hinge, job 2842288) Gaze converged to an extreme corner of image space: mean $(0.001, 0.998)$ with std $[0.020, 0.023]$ — well below the target std of 0.1 and therefore with the hinge active throughout. Task SPL plateaued at 0.05 as in pilot 1. The policy apparently found the minimum-task-cost gaze that partially satisfies the diversity pressure: a sigmoid saturated at one corner, producing a constant image-corner view.

Interpretation Both pilots argue that in our deterministic sigmoid-MLP gaze architecture with slow-gaze rollouts, gaze collapse is a pre-requisite for task learning rather than an accident. Task signal alone does not spontaneously produce informative fixation, and an exogenous anti-collapse regulariser moves gaze toward whatever region reduces regulariser loss fastest, which is generically *not* the task-useful region. A genuine H3 test likely requires (i) a stochastic gaze location with entropy bonus, so diversity is sampled at rollout time rather than enforced on a deterministic output; or (ii) a task-aligned intrinsic reward (e.g., information gain about goal-relative position); or (iii) an architectural change separating gaze generation from visual-feature extraction (e.g., attention-based gaze).

C Training Stability: Silent Weight Corruption in DD-PPO

During an initial foveated training run we observed a failure mode that did not manifest as a crash but as slow, silent corruption of the learned weights. After ≈170M frames of stable progress (reward ≈ 10, SPL ≈ 0.83), one mini-batch on one DDP worker produced a NaN gradient. Because `torch.nn.utils.clip_grad_norm_` does not sanitise non-finite gradients—the global norm becomes NaN, the clip coefficient becomes NaN, and `grad *= NaN` remains NaN—the subsequent `optimizer.step()` wrote NaN into every parameter. DDP all-reduce propagated the corruption across workers. The training loop did not raise: it continued for 2.5 days with reward=NaN and SPL$= 0$ until `torch.multinomial` finally rejected the NaN probability tensor. All checkpoints saved during this window were unrecoverable (90 of 97 parameter tensors entirely NaN).

To isolate the origin of the NaN gradient we re-ran training from the last clean checkpoint with `torch.autograd.set_detect_anomaly(True)` and per-module forward-output hooks. The anomaly traceback pinned the failure to `MseLossBackward0`, i.e. the backward pass of the value-head loss $\frac{1}{2}(V(s) - R)^2$; rare return-vs-value discrepancies overflowed float32 in the squaring step, producing non-finite gradients downstream.

Our mitigation is a minimally invasive patch to `PPO.before_step` that zeroes any non-finite gradient element before clipping. A bad mini-batch becomes a no-op update rather than a catastrophic one. The patch is a no-op on clean training paths. Across the five conditions reported here, no NaN events were observed during full training after deployment, and all final checkpoints contain only finite weights. We reported the underlying issue upstream (habitat-lab issue #2226) with a minimal reproduction and a proposed native fix.

D Foveation experiments: status and design

This appendix documents the four controlled foveation experiments described in §4.4. All four were submitted to the cluster at the time of writing; the table below tracks their state. Numbers will populate the corresponding paragraphs in §4.4 in revision.

Table 4: Foveation experiment status (submission time).

Experiment	Tests	Status
F1–F4 strength sweep ($\sigma_{\max} \in \{2, 4, 12, 20\}$)	encoder–memory race as continuous lever	in training
F3 log-polar foveation	spatial-sampling vs. blur bottleneck	in training
Encoder feature-map probe (matched / uniform / foveated)	where position information is held	in probing
Foveated-shifted (gaze $(0.49, 0.62)$)	gaze-location axis (links to H3)	in training

Why disclose this rather than wait. The four experiments are designed to falsify or refine specific predictions of the encoder–memory race framework, and each has a falsifiable signature. We disclose the design openly so that (i) the present submission is honest about what foveation-specific evidence already exists vs. what is in flight, and (ii) any reader can identify which result would change the paper's interpretation. The current evidence already places foveation-fixed and foveation-learned within the framework; the experiments above sharpen the "foveation" part of the story specifically.

E Encoder-capacity scaling sweep

[TODO: The scaling sweep trains matched-compute variants at input resolutions $\{32, 48, 64, 96, 128, 192\}$ alongside the existing blind (no encoder) and uniform 256×256 anchors. Probe results on the trained checkpoints will populate Figure 11 and this appendix section. Hypothesis: GPS probe R^2 decreases monotonically with input resolution, from $R^2 \approx 0.95$ at no-encoder to $R^2 \approx 0$ at 256×256, crossing into the pass-through regime between 48 and 128 pixels.]

Figure 11: Encoder-capacity scaling curve. **[TODO: Pending data.]** x-axis: input resolution in pixels; y-axis: GPS and compass probe R^2 (5-fold CV, mean$\pm\sigma$).